

Combinatorial Algorithms (Math 482), 2026 Spring Problem Set 2

Due by Wednesday, February 4, 2026.

1. **Towers of Hanoi.** Suppose that the three pins are labeled A , B , and C , and we need to move n disks from pin A to pin C . We know that this can be done in $2^n - 1$ moves.

Consider the following variant of the puzzle: We can move one disk between pins A and B , or between pins B and C (but we cannot move a disk directly between A and C). Design an algorithm that begins with n disks on pin A and ends with n disks on C . How many moves does your algorithm take?

2. Consider the following problem:

PARTITION($[A[1..n], p]$). We are given an array $A[1..n]$ and a number p (*pivot*). Rearrange the elements of the array $A[1..n]$ into a new array $B[1..n]$ such that $B[1], \dots, B[k] \leq p$ for some k and $B[k+1], \dots, B[n] > p$.

- (a) Design an algorithm, and write a pseudo-code for your algorithm.
 - (b) What is the running time of your algorithm? Express the running time as a function of n , using the Big-Oh notation.
3. In class, we discussed a recursive algorithm for the SELECTION problem.

SELECTION($A[1..n], k$). We are given an array $A[1..n]$ and a positive integer k . Find an element of $A[1..n]$ of rank k .

We divided the array $A[1..n]$ into subarrays of size 5, computed the median of each subarray, used the median-of-medians as a pivot, and recursed on the subarray on the left or on the right of the pivot. This strategy leads to the recurrence

$$T(n) = T\left(\frac{n}{5}\right) + T\left(\frac{7n}{10}\right) + O(n).$$

- (a) What recurrence would we obtain if we divide $A[1..n]$ into subarrays of size 3?
 - (b) What recurrence would we obtain if we divide $A[1..n]$ into subarrays of size $2k+1$ for some positive integer k ?
4. **Intersection Detection.** We are given n intervals $(A[1], B[1]), \dots, (A[n], B[n])$. Intervals $(A[i], B[i])$ and $(A[j], B[j])$ *overlap* if there is a real number x such that $A[i] < x < B[i]$ and $A[j] < x < B[j]$. Design a recursive algorithm that decides whether there are any two intervals that overlap, and runs in $O(n \log n)$ time. Describe the algorithm and analyze its running time using a recursion formula.